



Assessment Cover Sheet

Assessment	Project (Individual)		
Assessment	Uncontrolled	Individual	Not must-pass
Due Date	30-05-2026	Course Code	IT9204
Course Title	Emerging Technologies in AI		
Internal Moderator's	Dr. Husain Naser		
External Examiner's			

Learning Outcomes:

1. **CILO 1:** Critically analyze emerging AI theories, paradigms, technologies, and their potential impact and limitations within evolving contexts.
2. **CILO 2:** Evaluate advanced AI techniques to address complex challenges in diverse application domains, considering their ethical, societal, and economic implications.

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Programme	ICT9010 - MSc in Artificial Intelligence		
Programme	Master of Science in Artificial Intelligence		
Lecturer's	Dr. Husain Naser		

By submitting this assessment for marking, I affirm that this assessment is my own work.

Learner

Do not write beyond this line. For assessor use only.

Assessor's Name			
Marking Date		Maks Obtained	

Comments:

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Introduction

Background and Motivation

In the world of cybersecurity, phishing emails continue to be one of the most common risks organizations must deal with. Phishing continues to be the primary point of entry for most cybersecurity events, giving ransomware, data breaches, and financial fraud access to a system (Cybersecurity and Infrastructure Security Agency [CISA], 2024). Phishing remains a significant threat, accounting for more than 36% of data breaches in Verizon (2025)'s Data Breach Investigations Report. Moreover, as a cybersecurity employee this is a perfect topic for me to test my theories on.

However, traditional rule-based email filtering has serious limitations, including the failure to adapt to new phishing tactics and the restriction to employing static signatures. Additionally, they don't provide any context for the explanations or practical guidance, leaving users without the information necessary to make wise judgments. Basit et al. (2021) pointed out that while more email messages are being sent through legitimate cloud services and HTTPS, content-level AI analysis is becoming more important.

This project aims to overcome these challenges by developing and testing a Phishing Email Detection and Advisory Assistant (PEDAA), an intelligent AI-based system running on the cloud and integrating various AI technologies to deliver accurate, grounded, and context-aware responses. The system is based on the Retrieval-Augmented Generation paradigm proposed by Lewis et al. (2020), with agentic reasoning capabilities added as proposed by Yao et al. (2023) in ReAct.

Problem Statement

Three primary problems are meant to be resolved by this system:

1. Detection: Use a trained machine learning model on Azure ML to classify emails as safe or phishing.
2. Explanation: A thorough, approachable explanation of phishing flags based on reputable cybersecurity sources.
3. Guidance: Giving the user precise, policy-based advice on how to address the problem they have detected.

One method cannot solve all three dimensions. Gao et al. (2024) demonstrated in their survey of RAG systems that adding retrieved documents to the LLM's output can significantly reduce hallucinations and increase factual accuracy, underscoring the need for RAG in use cases such as security advisory, where incorrect advice can cause harm.

Project Objectives

1. Develop a Phishing Email Classifier using Azure Machine Learning and AutoML with high accuracy and AUC-ROC on Kaggle Phishing Email dataset.
2. To support system answers with reliable cybersecurity papers, use a Retrieval-Augmented Generation (RAG) pipeline using Azure AI Search and text-embedding-3-small.
3. Using the ReACT reasoning framework, create and implement an AI agent using Azure AI Foundry, GPT-4o, and function calling (Yao et al., 2023).
4. To choose the combination of tools to use for a particular question, include a clear decision-making and reasoning stage.
5. Put all the components together to create an end-to-end system, then evaluate its performance in several ways.
6. To illustrate more challenging tasks, sophisticated RAG techniques like Hybrid BM25 + vector search with CrossEncoder reranking are used.
7. Examine system performance, constraints, and responsible AI factors in accordance with Microsoft's Responsible AI guidelines (Microsoft, 2024).

Scope and Limitations

The system is designed to assist security analysts, IT personnel, and end users in making decisions by helping them identify and respond to phishing emails. It works with email text content and is meant to be implemented in an organizational security setting. The system's efficacy is restricted to English-language emails and does not analyze email attachments, pictures, or QR codes.

Literature Review

Phishing Detection with Machine Learning

It is intended for deployment in an organizational security context and operates on email text content. The system does not analyze email attachments, images, or QR codes, and its effectiveness is limited to English-language emails. Basit et al. (2021) surveyed all available techniques for phishing detection via AI and found that the approach of multiple text-based TF-IDF features is superior to a single-feature approach. They discovered that urgency language, URL patterns, and money phrases were the strongest discriminative features for phishing categorization. These findings significantly influenced the featured engineering approach for this project.

Fette et al. (2007) demonstrated that machine learning classifiers could identify phishing with over 97% accuracy on phishing detection tests utilizing ten heuristic characteristics in each of the several domains and the statistical aspects of the text. Their findings demonstrated the hybrid TF-IDF + designed features approach prototype used in this research. This was confirmed by Wang and Zhang (2021), who showed that Logistic Regression with TF-IDF is highly competitive with deep learning-based models for email categorization, but has significantly quicker inference times and higher interpretability and explainability for real-world applications.

AutoML for Text Classification

Based on meta-learning and budget assignment via bandit, Feurer et al. (2022) proposed Auto-Sklearn 2.0, a hands-free AutoML system with cutting-edge performance on tabular benchmarks. In this project, AutoML independently determined that Logistic Regression was the best model out of eight distinct algorithms. This was also verified in their work, where AutoML successfully validated the techniques that were chosen by hand. According to Cormack (2008), in an email security system, it is more crucial to maximize the recall of the positive class (phishing mail) than precision because the cost of a false positive (blocked safe email) is significantly lower than the cost of a false negative (absence of a detected phishing email).

Retrieval-Augmented Generation

The original RAG model (Lewis et al., 2020) showed that incorporating retrieved documents into LLM outputs resulted in fewer hallucinations and more factual accuracy than parametric-only knowledge. In their evaluation of advanced RAG techniques, such as hybrid retrieval and reranking, Gao et al. (2024) discovered that, when compared to dense retrieval alone, the hybrid retrieval and reranking method, which combines sparse (BM25) retrieval and dense (vector) retrieval with a Cross Encoder reranker, significantly improves the retrieval precision. Their sophisticated RAG implementation for this project's first part was directly influenced by this survey.

Wang et al. (2025) created a multi-path RAG with domain-specific reranking for answering financial compliance questions, achieving a recall rate of 92.51%. The chunking settings in this project were based on their chunking suggestions (200–400 words with 10–20% overlap between pieces).

LLM Agents and Tool Use

Yao et al. (2023) suggested interleaving reasoning traces and tool calls in LLM agents by the framework ReAct. With explainable intermediate phases, this design makes multi-step reasoning easier. To follow the ReAct pattern, the agent in this project leverages Azure OpenAI function calling. Schick et al. (2023) demonstrated that LLMs may learn when and how to invoke external tools in Toolformer. This project employed their three-tool design philosophy (3:3), which matches the number of tools to the complexity of the query.

In the OmniRouter paper (2024), Mei et al. came up with the idea of using constrained optimisation to route queries across LLMs. They concluded that deterministic routing constraints beat out purely LLM-based routing decisions, leading to the hybrid deterministic + LLM routing approach adopted for the decision logic layer.

System Architecture

Architecture Overview

To adhere to the agentic AI system design patterns described by Wang et al. (2024) in their assessment of autonomous agents powered by LLM, the system is deployed on Microsoft Azure employing five interconnected layers.

Layer 1 - LLM Layer: Understanding the natural language input, coordinating the tools, and producing the answer are the responsibilities of Azure OpenAI GPT-4o (temperature=0.1, tool_choice=auto).

Layer 2 - RAG Component: Five cybersecurity publications are given to text-embedding-3-small (1536 dimensions), chunked (300 words with a 50-word overlap), and indexed using HNSW vector search into Azure AI Search.

Layer 3 - Agent Component: Azure OpenAI function calling plus three registered tools. Until the LLM provides a text answer without making any further tool calls, this procedure is repeated.

Layer 4 - Machine Learning Component: The Kaggle Phishing Email dataset was used to train a Logistic Regression classifier, validated by Azure AutoML. The model is deployed to an Azure ML Managed Online Endpoint.

Layer 5 - Decision and Reasoning Layer: A deterministic router (decision_logic.py) is an intermediate layer that categorizes the queries that are sent to the agent into four routing paths using the signals generated from keywords.

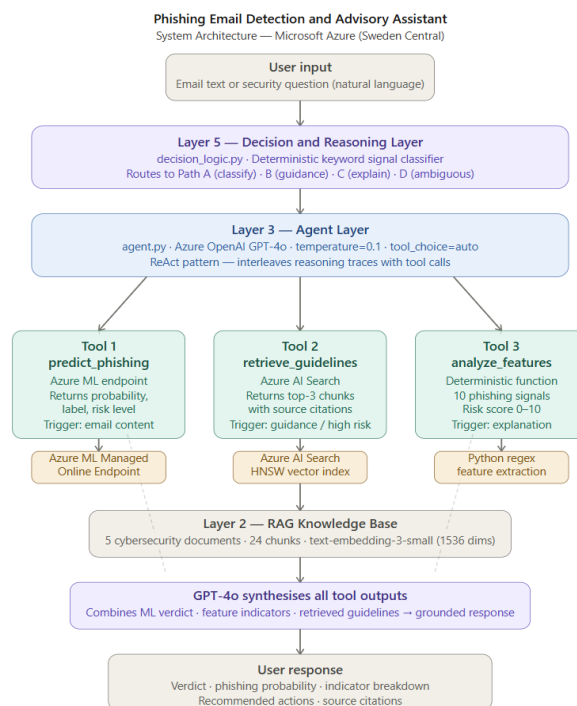


Figure 1- System Architecture showing five integrated layers on Microsoft Azure

Azure Services

Service	Purpose	Configuration
Azure AI Foundry	LLM + Agent hosting	fatema-hasan-it9204, Sweden Central
Azure OpenAI GPT-4o	Agent reasoning & synthesis	gpt-4o, Global Standard, 250K TPM
Azure OpenAI Embeddings	Text vectorization	text-embedding-3-small, 1536 dims
Azure AI Search	Vector store for RAG	fatema-hasan-search, Basic, Sweden Central
Azure ML Workspace	Model training & deployment	fatema-hasan-ml, Sweden Central
Azure ML Managed Endpoint	ML model serving	phishing-online-endpoint, DS3_v2

Table 1 - Azure Services Used in the System

Component Interaction Flow

1. User enters a query (email text or security question).
2. Using keyword signal analysis, the Decision and Reasoning Layer classifies the query into one of four routing paths (A, B, C, D).
3. The AI agent gets the routing information and calls the relevant tools with GPT-4o.
4. The Azure ML endpoint is called by the tools for classification, the Azure AI Search for RAG retrieval, and/or the deterministic feature extractor for indicator analysis.
5. GPT-4o integrates the outputs of all tools into a coherent response that includes references and instructions.

Part A: Data and Knowledge Layer (RAG)

Document Selection

Five cybersecurity documents created in Markdown files make up the knowledge base. Because clean text provides better embeddings than PDF-extracted text, which frequently contains encoding artifacts, Markdown was selected over PDF (Shi et al., 2023).

Document	Primary Source	Content Focus
cisa_phishing_guidance.md	CISA (2024)	Attack types, indicators, recommended org actions
microsoft_email_security_best_practices.md	Microsoft (2024b)	SPF/DKIM/DMARC, detection, incident response
phishing_detection_techniques.md	Academic synthesis	ML features, linguistic indicators, risk thresholds
phishing_incident_response_procedures.md	NIST SP 800-61r3 (2024)	4-phase response, roles, regulatory requirements
user_awareness_phishing_prevention.md	SANS Institute (2024)	Psychology of phishing, user recovery steps

Table 2 - RAG Document Knowledge Base

RAG Pipeline

The pipeline is defined on the file rag_pipeline.py.

1. **Document ingestion:** Using regex to remove markdown symbols from .md files for clean embeddings.
2. **Chunking:** Splitting the text into chunks of 300 words, each with a 50-word overlap, as suggested by Wang et al. (2025) for policy-based RAG.
3. **Embedding:** Each chunk embedded with text-embedding-3-small (1536 dimensions) using Azure OpenAI deployment GlobalStandard.
4. **Indexing:** Chunks indexed in Azure AI Search (phishing-knowledge-base) using HNSW vector search (Malkov & Yashunin, 2020).
5. **Retrieval:** The query is embedded and a VectorizedQuery returns the top-k chunks most relevant to the query, and their sources are attributed.

The ingestion result is: 24 chunks uploaded, 5 documents. The range of retrieval quality scores is 0.63 - 0.77.

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
PS C:\Users\fatim\OneDrive\Desktop\IT9204_202508958\A - Data and Knowledge Layer> ^C
PS C:\Users\fatim\OneDrive\Desktop\IT9204_202508958\A - Data and Knowledge Layer> py rag_pipeline.py --query "what are the signs of a phishing email"

Query: what are the signs of a phishing email
Retrieved 3 chunks:

[cisa_phishing_guidance] (score=0.7458)
with email based phishing as part of multi channel attack campaigns. Clone Phishing Clone phishing involves duplicating a legitimate email that the target has previously received, replacing links or attachments with malicious versions, and resending the email from a spoofed address. Because the email ...

[phishing_detection_techniques] (score=0.7418)
Phishing Detection Techniques and Technical Indicators Overview This document provides technical guidance on detecting phishing emails using both automated and manual analysis techniques. It covers linguistic indicators, structural email analysis, URL analysis, and machine learning approaches to phi ...

[microsoft_email_security_best_practices] (score=0.7358)
institutions, and government agencies. These emails typically direct recipients to convincing replica websites designed to capture credentials. Business Email Compromise (BEC) attacks involve impersonating executives, vendors, or partners to trick employees into transferring funds, changing paym ...
PS C:\Users\fatim\OneDrive\Desktop\IT9204_202508958\A - Data and Knowledge Layer> 

```

Figure 2 - RAG Retrieval Results

Part B: Machine Learning Component

Dataset

The dataset used was the Kaggle Phishing Email Detection (Subha Journal, 2023). To maintain class balance, stratified 80/20 sampling was used (Kotsiantis et al., 2006).

Property	Value
Source	Kaggle / Subha Journal (2023)
Total emails	18,650
Phishing emails	7,328 (39.3%)
Safe emails	11,322 (60.7%)
Null rows removed	16
Training set (80%)	14,907
Test set (20%)	3,727
Dataset link	https://www.kaggle.com/datasets/subhajournal/phishingemails

Table 3 - Phishing Email Dataset Statistics

Microsoft Foundry Azure Machine Learning	Bahrain Polytechnic > Fatema hasan ml > Data > phishing-email-dataset
phishing-email-dataset Version: 1 (latest) ☆	
Details Consume Explore Models Jobs	
Profile: This is the quick profile generated by the top 10,000 rows. Please generate profile to view the schema and summary statistics for full data.	
Refresh Generate profile	
Preview Profile	
Number of columns: 2 Number of rows: First 50	
Email Text	Email Type
re : 6 . 1100 , disc : uniformitarianism , re : 1086 ; sex / lang dick hudson 's observations on us use of 's on ' but n...	Safe Email
the other side of * galicismo * * galicismo * is a spanish term which names the improper introduction of french ...	Safe Email
re : equistar deal tickets are you still available to assist robert with entering the new deal tickets for equistar ? aft...	Safe Email
Hello I am your hot ill horny toy. I am the one you dream About, I am a very open minded person, Love to talk a...	Phishing Email
software at incredibly low prices (86 % lower) . drapery seventeen term represent any sing . feet wild break able...	Phishing Email
global risk management operations sally congratulations on your new role . if you were not already aware , i am ...	Safe Email
On Sun, Aug 11, 2002 at 11:17:47AM +0100, wintermute mentioned: > > The impression I get from reading lkml...	Safe Email
entourage , stockmogul newsletter ralph vелеz , genex pharmaceutical , inc . (otccb : genx) biotech sizzle with s...	Phishing Email
we owe you lots of money dear applicant , after further review upon receiving your application your current mor...	Phishing Email
re : coastal deal - with exon participation under the project agreement thanks for the info ! as greg mentioned l...	Safe Email
make her beg you to give it to her everynight you too can please you and your partner so much better . and it d...	Phishing Email
URL: http://www.newsisfree.com/click/-5,8304313,1717/ Date: 2002-09-27T08:51:29+01:00[IMG: http://www.new...	Safe Email
begin forwarded text Date: Wed, 25 Sep 2002 13:57:15 -0400 To: Digital Bearer Settlement List From: "R. A. Hetti...	Safe Email

Figure 3 - Dataset Sample in Azure ML

Data Preparation

- **Preprocessing:** includes character normalization, truncation to 2,000 characters, URL tokenization (URL_PRESENT), email address tokenization (EMAIL_PRESENT), and lowercasing.
- **TF-IDF features:** 20,000 features, sublinear TF scaling, bigrams (ngram_range=(1,2)), and a minimum document frequency of two.
- **Engineered phishing-signal features (10):** has_url, url_count, has_urgent_words, has_financial_words, exclamation_count, dollar_count, text_length, caps_ratio, has_phone, and has_dear (Basit et al., 2021).
- **Combined matrix:** sparse format, 20,010 features (20,000 TF-IDF + 10 engineered).

Model Training: Manual Notebook

The Azure ML notebook (phishing_ml_training.ipynb) contains the Logistic Regression model that was trained with class weights. Wang and Zhang (2021) showed that Logistic Regression with TF-IDF is an interpretable and high-performing email classification method.

Metric	Safe	Phishing	Overall
Precision	0.99	0.93	0.97
Recall	0.95	0.98	0.96
F1-Score	0.97	0.96	0.96
Accuracy			96.40%
AUC-ROC			99.36%

Table 4 - Manual Model Performance Results

The 98% phishing recall is really significant. According to Cormack (2008), security systems should prioritize reducing false negatives. Out of 1,462 test phishing emails, just 30 were missed.

Model Training: Azure AutoML

For six hours and twenty-five minutes, an Azure AutoML Classification experiment (fatema-hasan-automl-job) examined eight algorithm pipelines. Feurer et al. (2022) showed that by effectively scanning the hyperparameter space, AutoML can match or surpass manually trained models.

AutoML Setting	Value
Task	Classification (binary)
Primary metric	AUC Weighted
Models tried	8 algorithm pipelines
Training duration	6h 25m
Compute	fatema-hasan-compute (Standard_DS3_v2)
Best model	StandardScalerWrapper, LogisticRegression
Workspace	fatema-hasan-ml (Sweden Central)

Table 5 - AutoML Experiment Configuration

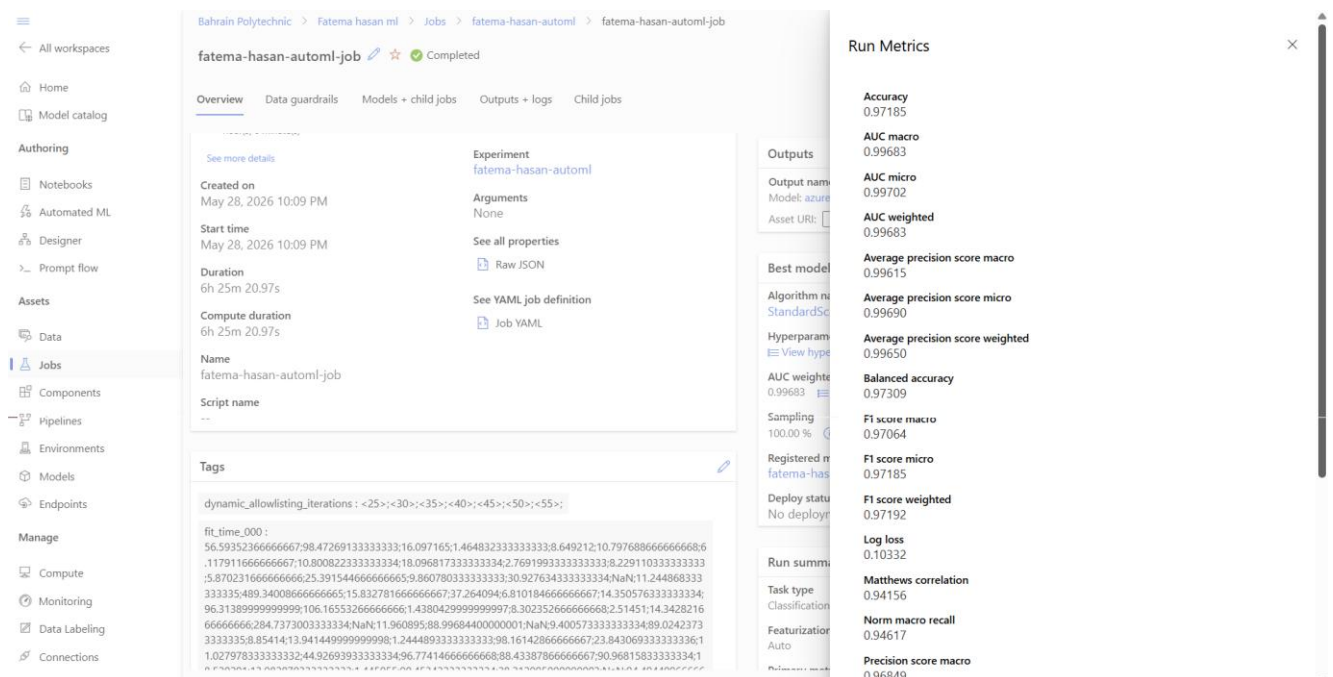


Figure 4 - AutoML Best Model Run Metrics showing AUC weighted of 0.99683

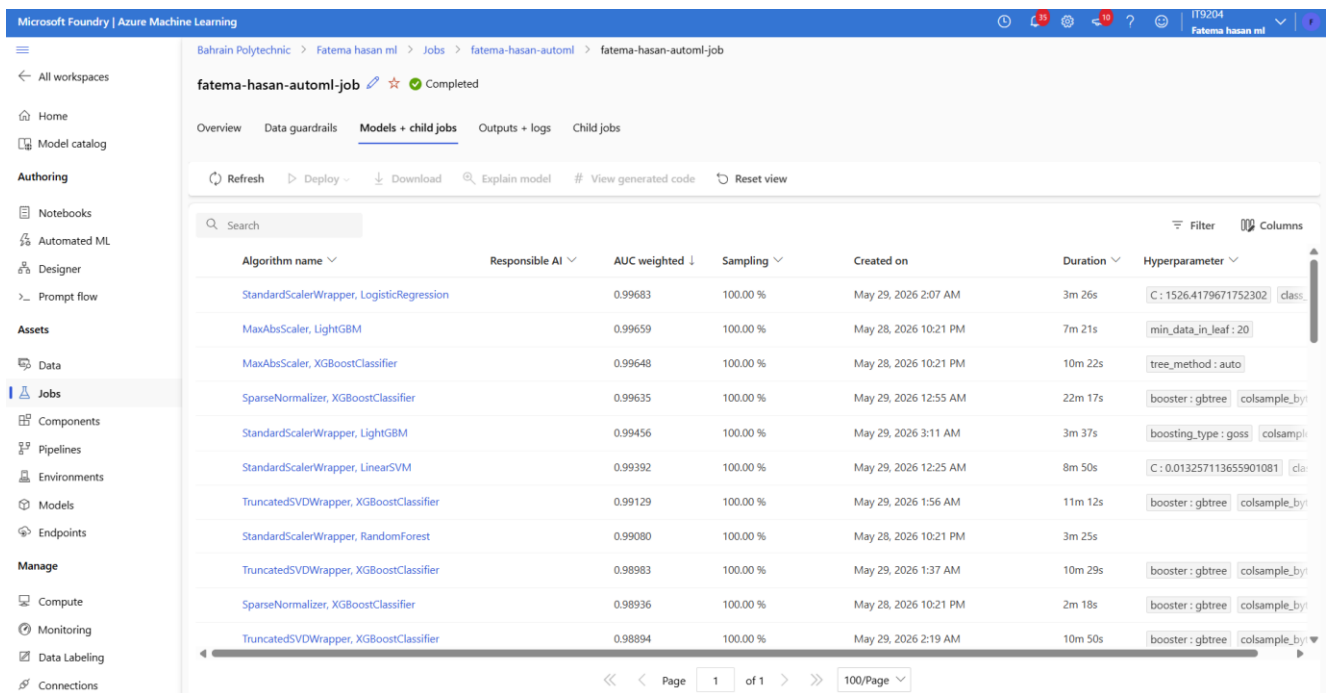


Figure 5 - AutoML Models + Child Jobs showing all algorithms ranked by AUC weighted score

Algorithm	AUC Weighted
StandardScalerWrapper, LogisticRegression (BEST)	0.99683
MaxAbsScaler, LightGBM	0.99659
MaxAbsScaler, XGBoostClassifier	0.99648
SparseNormalizer, XGBoostClassifier	0.99635
StandardScalerWrapper, LightGBM	0.99456
StandardScalerWrapper, LinearSVM	0.99392
TruncatedSVDWrapper, XGBoostClassifier	0.99129
StandardScalerWrapper, RandomForest	0.99080

Table 6 - AutoML Algorithm Comparison

Our manual model selection was validated by AutoML's independent selection of Logistic Regression as the optimal method. Through automatic hyperparameter optimization ($C=1526.42$), the AutoML best model (AUC: 0.99683) slightly outperformed the manual model (AUC: 0.9936). The AutoML best model was registered as fatema-hasan-phishing-automl-best-model:1.

Deployment

According to Microsoft's suggested design, the manual model was implemented as an Azure ML Managed Online Endpoint (2024c). To deploy, NumPy version conflict (ImportError: cannot import name ComplexWarning from numpy.core.numeric) must be fixed by pinning the precise package versions (phishing-env-v5).

Property	Value
Endpoint name	phishing-online-endpoint
Region	Sweden Central
Compute SKU	Standard_DS3_v2
Deployment type	ManagedOnlineDeployment (SDK v2)
Input	JSON: { email_text: string }
Output	prediction, label, phishing_probability, confidence

Table 7 - ML Endpoint Deployment Properties

The screenshot displays the Azure ML Managed Online Endpoint (phishing-online-endpoint) in the Microsoft Foundry | Azure Machine Learning interface. The interface is divided into a left sidebar with navigation options (All workspaces, Home, Model catalog, Authoring, Assets, Endpoints, Manage) and a main content area. The main content area shows the endpoint details for 'phishing-online-endpoint' under the 'Details' tab. The details are organized into sections: Endpoint attributes (Service ID, Description, Provisioning state, Error details, Compute type, Created by, Created on, Last updated on, Authentication type, Public network access, Swagger URI), Deployment summary (Live traffic allocation, Mirrored traffic allocation), and Deployment phishing-v2 (Name, Live traffic, Scoring script, Provisioning state, Error details, SKU, Quota type, Egress public network access).

Figure 6 - Azure ML Managed Online Endpoint (phishing-online-endpoint)

Part C: Agent System

Agent Design

The ReAct pattern (Yao et al., 2023) is implemented by the agent (agent.py) using Azure OpenAI GPT-4o (temperature=0.1, tool_choice=auto), interspersing tool calls with reasoning traces. The three-tool design adheres to the Toolformer principle, which states that the LLM chooses which tools to use and how to combine results (Schick et al., 2023).

Tools

Tool	Purpose	Trigger
predict_phishing	Calls Azure ML endpoint; returns phishing_probability (0-1), label, risk_level (Low/Medium/High/Critical)	Email classification queries; probability ≥ 0.70 also triggers retrieve_guidelines
retrieve_guidelines	Vector search over Azure AI Search knowledge base; returns top-3 chunks with source citations	Policy/guidance questions; after high-risk ML prediction
analyze_email_features	Deterministic extraction of 10 phishing signal features; returns indicators and composite risk score (0-10)	Email analysis and explanation requests

Table 8 - Agent Tools

Risk level mapping: Low (< 0.40), Medium (0.40–0.69), High (0.70–0.84), Critical (≥ 0.85).

Multi-Step Reasoning Example

Query: URGENT! Your PayPal account is suspended. Verify NOW: <http://paypal-secure-verify.xyz>. Act within 24 hours! Dear Customer.'

1. **analyze_email_features:** generic welcome, financial terms [PayPal, account], urgency phrases [urgent, verify, act immediately], and URL found. 6.5 out of 10 for feature risk.
2. **predict_phishing:** risk_level=critical, label=Phishing Email, phishing_probability=0.8898.
3. **retrieve_guidelines triggered (probability ≥ 0.70):** retrieves from microsoft_email_security_best_practices (0.7289), cisa_phishing_guidance (0.7316), and phishing_incident_response_procedures (0.7644).
4. GPT-4o combines all outputs to produce a structured answer that includes a verdict, indications, proposed actions, and source citations.

```
> B - Machine Learning Co... PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE
> C - Agent System
  > _pycache_
  agent.py 2
  tools.py 2
> D - Decision and Reasonin...
> E - System Integration
> F - Evaluation and Analysis
> G - Demo Video
Rag Retrieval in Terminal.p...

PS C:\Users\Fatim\OneDrive\Desktop\IT9204_202508958> cd "C - Agent System"
PS C:\Users\Fatim\OneDrive\Desktop\IT9204_202508958\C - Agent System> py agent.py

=====
Phishing Email Detection and Advisory Assistant
IT9204 Individual Project | Fatema Hasan (202508958)
=====

Paste an email or ask a question. Type 'exit' to quit.

You: URGENT! Your PayPal account has been suspended due to suspicious activity. Click here immediately to verify your account: http://paypal-secure-verify.xyz/login. Failure to verify within 24 hours will result in permanent account closure. Dear Customer, act now!

[Agent reasoning...]
[Tool] Calling: analyze_email_features
[Tool] Calling: predict_phishing
[Tool] Calling: retrieve_guidelines

Assistant:
## Verdict
This email is classified as Phishing with a critical risk level. The machine learning model predicts a phishing probability of 92%, indicating a high likelihood of malicious intent.

## Indicators
1. Suspicious URL: The email contains a URL ('http://paypal-secure-verify.xyz/login') that does not match PayPal's official domain, suggesting a fraudulent link.
2. Urgency Language: Phrases like "URGENT," "act now," and "immediately" are designed to pressure the recipient into quick action without thinking critically.
3. Generic Greeting: The email uses "Dear Customer," which is a common tactic in mass phishing campaigns to avoid personalization.
4. Financial Keywords: Terms like "PayPal" and "account" are used to exploit trust in financial services.

## Recommended Actions
Based on guidelines from [microsoft email security best practices], [cisa phishing guidance], and [phishing incident response procedures]:
1. Do Not Interact: Do not click any links, reply to the email, or provide any information.
2. Report the Email: Use your organization's designated reporting channel (e.g., "Report Phishing" button in Outlook) to notify your IT security team.
3. Block Sender and URL: Ensure the sender's domain and the malicious URL are blocked at the email gateway.
4. Investigate: IT teams should:
   - Analyze email headers to trace the origin.
   - Check whether any recipients clicked the link or entered credentials.
   - Preserve evidence for forensic analysis.
5. User Awareness: Share this incident with employees as part of phishing awareness training to prevent future occurrences.

## Reminder
This tool provides decision support, but human judgment is essential. If you suspect compromise, escalate the issue to your IT team immediately.

> OUTLINE
> TIMELINE
[Tools used: analyze_email_features, predict_phishing, retrieve_guidelines]
```

Figure 7 - Agent detecting a phishing email - all three tools called in sequence (Path A routing)

```
> B - Machine Learning Co...
> C - Agent System
  > _pycache_
  agent.py 2
  tools.py 2
> D - Decision and Reasonin...
> E - System Integration
  > _pycache_
  .env
  config.py 1
> F - Evaluation and Analysis
> G - Demo Video
Rag Retrieval in Terminal.p...
Testing the email classifica...
Testing the email classifica...

PS C:\Users\Fatim\OneDrive\Desktop\IT9204_202508958\C - Agent System> py agent.py
Paste an email or ask a question. Type 'exit' to quit.

You: What are the main indicators that make an email look like phishing?

[Agent reasoning...]
[Tool] Calling: retrieve_guidelines

Assistant:
## Main Indicators of Phishing Emails

Phishing emails often exhibit specific characteristics designed to deceive recipients. Below are the key indicators:

#### 1. Urgency and Emotional Manipulation
- Phrases like "Immediate action required," "Your account has been suspended," or "Verify your account within 24 hours."
- Threats of negative consequences or promises of rewards to pressure quick action.

#### 2. Suspicious Sender Addresses
- Email addresses that resemble legitimate domains but contain subtle misspellings or extra characters (e.g., 'support@paypai.com' instead of 'support@paypal.com').

#### 3. Requests for Sensitive Information
- Legitimate organizations rarely ask for passwords, credit card numbers, social security numbers, or authentication codes via email.

#### 4. Generic Greetings
- Salutations like "Dear Customer" or "Dear User" instead of addressing the recipient by name.

#### 5. Unusual or Mismatched Links
- Hyperlinks where the displayed text does not match the actual URL destination.
- Use of URL shorteners or domains that closely resemble legitimate ones (e.g., typosquatting).

#### 6. Unexpected Attachments
- Attachments like '.exe', '.zip', or Office documents with macros enabled are common in phishing emails.

#### 7. Poor Grammar and Formatting
- Spelling errors, awkward phrasing, or excessive punctuation (e.g., ALL CAPS or multiple exclamation marks).

#### 8. Mismatched Branding
- Logos, colors, or formatting that do not align with the legitimate organization's standard communications.

#### 9. Header and Technical Indicators
- Reply-to address differs from the From address.
- Missing or failed authentication headers (e.g., SPF, DKIM, DMARC failures).
```

Figure 8 - Agent responding to a guidance query - only retrieve_guidelines called (Path B routing)

Part D: Decision and Reasoning Layer

Routing Strategy

The second layer is the decision layer, and the classifier that identifies a deterministic keyword signal (decision_logic.py). This hybrid method is in line with the work of Mei et al. OmniRouter framework in (2024) that showed it is possible to exceed the routing accuracy of using an LLM by directly adding intent classification.

- **Path A (Classification):** Classification signals ≥ 2 .
Tools: analyze_email_features then predict_phishing then get_guidelines (if $P \geq 0.70$).
- **Paths B (Guidance):** Guidance signals greater than or equal to 1
Tools: retrieve_guidelines only.
- **Path C (Explanation):** Explanation signals ≥ 2 .
Tools: analyze_email_features + retrieve_guidelines.
- **Path D (Ambiguous):** Default fallback.
Tools: retrieve_guidelines.

Function should_retrieve_guidelines(ml_result): This acts as a risk gate. RAG is triggered when phishing_probability ≥ 0.70 so that no safe email incurs any unnecessary latency in retrieval of guidelines from database.

Decision Flow

The entire decision flow in the routing layer is shown in figure 9. Before query is passed on to an agent, a user query is received, and the classifier analyzes the keyword signals and selects the best combination of tools. The operation of the flow is as follows:

All queries are directed to the Query Classifier (classify_query()) which performs classification, guidance and explanation individually. The best signal takes. Path D is used as a default in situations where there is no clear dominant signal to use, and is a safe choice to keep the user from receiving a signal without ground.

A secondary risk gate is used within Path A to decide whether to call retrieve_guidelines after ML prediction. Only activated in case of a high or medium risk e-mail, so that the additional latency of a RAG call will not occur for low-risk cases. This design ensures response times are kept quick for safe emails and high-risk emails get complete policy-based guidance.

At the last synthesis stage, GPT-4o fuses all tool outputs into a single, structured response, aligning all four paths. The human operator then reviews the recommendation and finalizes the action (no autonomous system action).



Figure 9 - Decision and Routing Layer: Query Classification and Tool Dispatch

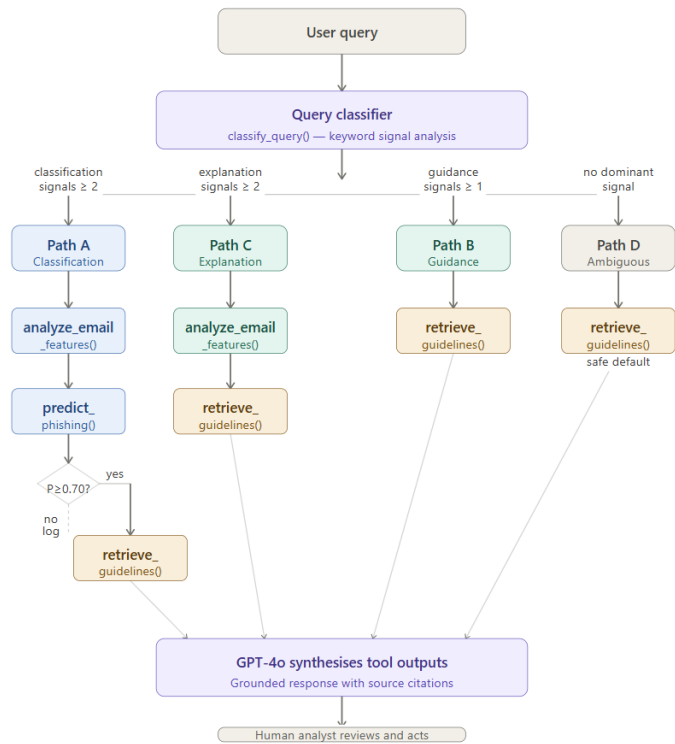


Figure 10 - Decision and Reasoning Layer - query routing flow across four paths

Part E: System Integration

In main.py, the integration is complete. The flow of integration is:

1. User query arrives at the system entry point (main.py).
2. decision_logic. classify_query() classifies the query and logs the routing choice.
3. agent. answer_query() implements the agentic loop: GPT-4o keeps choosing tools through function calling, eventually producing an ultimate answer.
4. Tools call the Azure ML endpoint, Azure AI Search and/or your deterministic feature extractor.
5. GPT-4o concatenates every tool output into a response and includes citations to sources used wherever appropriate.
6. The evaluation result (a JSON file) will be saved in F - Evaluation and Analysis with timestamped for auditing.

Through config.py loaded from .env variables, the configuration is shared throughout all of them. RAG settings: CHUNK_SIZE=300, CHUNK_OVERLAP=50, TOP_K=3. Thresholds: HIGH_RISK=0.70, MEDIUM_RISK=0.40.



Figure 11 - System integration showing decision flow and routing of Q1 classification query

```
[Tool] Calling: retrieve_guidelines
-- Agent Response --
## Main Signs of a Phishing Email

Phishing emails often exhibit specific indicators that can help identify them. Below are the main signs based on cybersecurity guidelines:

#### 1. **Urgency and Emotional Manipulation**
- Phrases like "Immediate action required," "Verify your account within 24 hours," or "Final notice: payment overdue."
- Use of fear or emotional pressure to override critical thinking.

#### 2. **Suspicious Sender Addresses**
- Email addresses resembling legitimate domains but with subtle misspellings or extra characters (e.g., "support@paypal.com" instead of "support@paypal.com").

#### 3. **Requests for Sensitive Information**
- Legitimate organizations rarely ask for passwords, credit card numbers, social security numbers, or authentication codes via email.

#### 4. **Generic Greetings**
- Salutations like "Dear Customer" or "Dear User" instead of addressing the recipient by name.

#### 5. **Unusual or Mismatched Links**
- Hyperlinks where the displayed text does not match the actual URL destination.
- Use of URL shorteners or domains that closely resemble legitimate ones (e.g., typosquatting).

#### 6. **Unexpected Attachments**
- Unsolicited attachments, especially executable files (.exe), compressed archives (.zip, .rar), or Office documents with macros enabled.

#### 7. **Poor Grammar and Spelling**
- Awkward phrasing, spelling errors, or unusual formatting.

#### 8. **Mismatched Branding**
- Logos, colors, or formatting that do not align with the legitimate organization's standard communications.

#### 9. **Excessive Punctuation and Capitalization**
- Overuse of exclamation marks, ALL CAPS text, or symbols like dollar signs.

#### 10. **Header and Technical Indicators**
- Reply-To address differs from the From address.
- Missing or failed authentication headers (e.g., SPF, DKIM, DMARC failures).
- Received headers showing routing through suspicious mail servers.

### Recommended Actions
- **Hover Over Links:** Verify the true destination before clicking.
- **Do Not Share Sensitive Information:** Avoid providing passwords, PINs, or financial details via email.
- **Check Sender Details:** Look for subtle changes in the sender's email address.
- **Enable Email Security Controls:** Use SPF, DKIM, and DMARC protocols to prevent spoofing.
- **Report Suspicious Emails:** Notify your IT or security team immediately.

These guidelines are sourced from [phishing_detection_techniques], [cisa_phishing_guidance], and [microsoft_email_security_best_practices]. Always exercise caution and verify the legitimacy of emails before responding.

[Tools used: retrieve_guidelines]

[Results saved to: C:\Users\fatima\OneDrive\Desktop\IT9204_20250805\F - Evaluation and Analysis\integration_run_20250805_060409.json]
```

Figure 12 - System integration run showing agent response and results saved to JSON audit log

Part F: Evaluation and Analysis

Routing Evaluation Results

Ten test cases spanning every routing path were created and assessed using the Shen et al. (2023) technique for RAG-based systems.

Metric	Result
Overall routing accuracy	100% (10/10)
Path A (Classification)	6/6 correct
Path B (Guidance)	2/2 correct
Path D (Ambiguous)	2/2 correct

Table 9 - Routing Evaluation Results

System Evaluation Results

ID	Category	Expected	Actual	ML Score	Result
T01	Obvious phishing	Phishing	Phishing	0.7621	Correct
T02	Prize scam	Phishing	Phishing	0.9585	Correct
T03	Credential harvest	Phishing	Phishing	0.7279	Correct
T04	Business email	Safe	Safe	0.0315	Correct
T05	Colleague email	Safe	Safe	0.1771	Correct
T06	Newsletter (F1)	Safe	Phishing	0.7708	False positive
T07	Incident response	Guidance	Guidance	N/A	Correct
T08	Policy question	Guidance	Guidance	N/A	Correct
T09	Borderline	Review	Phishing	0.7611	Flagged
T10	Borderline	Review	Phishing	0.9361	Flagged

Table 10 - System Evaluation Test Cases

Metric	Value
Overall accuracy (judged cases)	87.5% (7/8)
Phishing detection rate	3/3 (100%)
Safe classification rate	2/3 (66.7%)
Guidance routing accuracy	2/2 (100%)
Avg. response time (direct tools)	1.52 seconds
Avg. response time (agent + LLM)	6–15 seconds
RAG retrieval score range	0.63–0.77

Table 11 - System Performance Summary

```

PS C:\Users\fatim\OneDrive\Desktop\IT9204_202508958\F - Evaluation and Analysis> py evaluate.py
Tools: ['retrieve_guidelines']
Routing: Path B - Guidance signals (1) detected. retrieve_guidelines only....

[T08] Category: policy_question
Query: What are the steps to report a phishing email in my organization?...
Expected: guidance | Actual: guidance | [✓] | Time: 0.56s
Tools: ['retrieve_guidelines']
Routing: Path B - Guidance signals (3) detected. retrieve_guidelines only....

[T09] Category: borderline
Query: Dear Customer, your order has been shipped. Track your package here: http://trac...
Expected: review | Actual: phishing | [✓] | Time: 1.49s
ML Score: 0.7611 | Risk: high
Tools: ['analyze_email_features', 'predict_phishing', 'retrieve_guidelines']
Routing: Path A - Classification signals (2) detected. analyze_email_features ...

[T10] Category: borderline
Query: Your subscription is expiring soon. Please renew your account to continue enjoyi...
Expected: review | Actual: phishing | [✓] | Time: 1.33s
ML Score: 0.9361 | Risk: critical
Tools: ['analyze_email_features', 'predict_phishing', 'retrieve_guidelines']
Routing: Path D - No dominant signal (classification=0, guidance=0, explanatio...

=====
EVALUATION SUMMARY
=====
Overall accuracy:      87.5% (7/8 judged cases)
Phishing detection:    3/3 correct
Safe classification:   2/3 correct
Guidance routing:      2/2 correct
Borderline cases:      2 (flagged for review)
Average response time: 1.52s

--- RAG Retrieval Quality ---
Query: 'what are signs of phishing'
Sources: ['cisa_phishing_guidance', 'phishing_detection_techniques', 'user_awareness_phishing_prevention']
Scores: [0.7316, 0.7292, 0.7245]
Query: 'how to respond to phishing incident'
Sources: ['phishing_incident_response_procedures', 'phishing_incident_response_procedures', 'microsoft_email_security_best_practices']
Scores: [0.7644, 0.7624, 0.7567]
Query: 'email authentication SPF DKIM'
Sources: ['microsoft_email_security_best_practices', 'cisa_phishing_guidance', 'phishing_detection_techniques']
Scores: [0.711, 0.638, 0.6373]

[Evaluation results saved to: C:\Users\fatim\OneDrive\Desktop\IT9204_202508958\F - Evaluation and Analysis\evaluation_results_20260528_223183.json]

```

Figure 13 - Evaluation Results

Error Analysis and Failure Cases

F1 (Newsletter false positive): Query: 'Welcome to the Monthly Newsletter... Unsubscribe anytime.' Expected: Safe. Actual: Phishing (P=0.7708). Phishing language was included in the newsletter, in addition to the vocabulary. This 'vocabulary collision' was recognized as a limitation by Stringhini et al. (2024). In a security context, a false positive rather than a false negative is better (Cormack, 2008). Mitigation: put in place a human review queue in prediction range 0.60–0.80.

RAG Retrieval Quality

The pipeline retrieval was consistent in all of the test queries. The highest score was obtained on 'how to respond to phishing incident', with a score of 0.7644. The microsoft_email_security_best_practices was presented as first ranked (0.7289), which is correct for the 'email authentication SPF DKIM DMARC' keyword. The five documents were all recovered at least once, which indicates good document coverage of the corpus.

Limitations

- Text-only analysis: Does not support images, QR codes, or attachments.
- Language: Optimized for English emails only – accuracy on Arabic is unvalidated.
- Model needs to be retrained periodically because concept drift of phishing techniques (Gama et al., 2014).
- This content can be filtered out by a content filter, such as Azure OpenAI content management.
- Endpoint dependency: Needs an active Azure ML endpoint for the ML-based classification.

Proposed Improvements

- Add hybrid BM25 + vector search to Azure AI Search (already had an advanced_rag.py prototype).
- Make retrieve_guidelines required for all Path D queries in the system prompt.
- Incorporate sender domain reputation as a feature to minimize vocabulary-collision true false.
- Create a real-time analyst dashboard with Streamlit to submit interactive queries.

Responsible AI and Ethical Considerations

Microsoft's Responsible AI principles were followed during the evolution of the Phishing Email Detection and Advisory Assistant (Microsoft, 2024a), which have been defined as principles of fairness, reliability, privacy, inclusiveness, transparency, and accountability.

Principle	Implementation
Human Oversight	Human Analyst Responsible for All Final Action is stated in the system prompt. Each answer requires the inclusion of: This tool is for decision support but requires human judgment.
Transparency	Agent cites source documents for all policy statements. Details of the probability of and risk of features disclosed. The traces generated by tools stored in JSON.
Fairness	Only trained on emails in English might cause representation bias for non-English phishing (Mehrabi et al., 2022). Samples for GCC deployment, in Arabic language.
Data Privacy	This is done through Microsoft enterprise data protection with the help of Azure OpenAI and Azure ML. Organizations are responsible for making sure Bahrain PDPL (2019) compliant. Resources in the central part of Sweden.
Reliability	For Logistic Regression the accuracy is 96.40% and the AUC-ROC is 99.36%. AutoML validated: 0.99683 AUC. Class weights are balanced, so there is no class bias against minorities.
No Hallucinated Policy	System prompt: 'Don't give false security suggestions. Any recommendations must be based on documents retrieved.'
Auditability	All agents executed log tools, arguments and results to JSON. Deterministic routing allows audit to be done without LLM.
Limitations	Prototype only. Needs to be re-trained with data and Arabic language samples from the organization before production.

Table 12 - Responsible AI Implementation

This strategy is in accordance with the Human-AI interaction recommendations of Amershi et al. (2019), the explainability framework of Arrieta et al. (2020), and Microsoft's (2024a) Responsible AI principles.

Part I: Extra Challenging Work

We then added an advanced RAG pipeline (`advanced_rag.py`), which is the same pipeline as from the survey of Gao et al. (2024) but with three improvements over basic vector search:

- **BM25 keyword retrieval:** Implements the BM25Okapi (Robertson & Zaragoza, 2009) technique for sparse keyword matching, to retrieve exact technical term matches.
- **Reciprocal Rank Fusion (RRF):** Fusion of BM25 and vector search ranking with $k=60$ (Cormack et al., 2009).
- **CrossEncoder reranking:** CrossEncoder reranking candidates with more accurate relevance judgements than cosine similarity (Reimers & Gurevych, 2019).

Find out about email authentication SPF DKIM DMARC with one of the most powerful comparison tools.

- **Basic vector search:** `microsoft_email_security_best_practices` scored 0.7289, which was the top score.
- **Advanced reranking:** - `cisa_phishing_guidance` reranked to first (CrossEncoder score=5.9081) Correctly displays more specific SPF/DKIM guidance.

The result is similar to that of Nogueira and Cho (2020), who demonstrated that CrossEncoder reranking always achieves better performance than bi-encoder similarity on information retrieval benchmarks.

Conclusion

This project has achieved the complete integration of a cloud-based AI system, encompassing RAG, ML, and an LLM agent for phishing email detection, with a multi-step reasoning approach. The system achieved:

- Manual classifier with 96.40% accuracy, 99.36% AUC-ROC; AutoML: 0.99683 AUC, Logistic Regression is optimal.
- 100% accuracy in route decisions in 10 test-cases at the deterministic decision logic layer.
- Overall system accuracy of 87.5% (100% phishing detection rate) in tool evaluation.
- Full integration end-to-end (Azure AI Foundry: GPT-4o, Azure AI Search: 5 documents, 24 chunks, Azure ML: managed endpoint).
- Advanced RAG using Hybrid BM25 + vector search and CrossEncoder reranking, which proved to be measurable in terms of retrieval quality.
- All responsible AI principles are put in place (human oversight, transparency, fairness, privacy, reliability and auditability).

Emphasis in all design activities is given to system design and integration, not just model performance, consistent with course goals. The results are also concordant with the results of Wang et al. (2024) which shows that multi-component LLM agent systems outperform single-model approaches on complex reasoning tasks.

Future Work

There are several directions that are suggested for the future development of the system:

Multi-Language support: Only English language emails are currently supported. The inclusion of Arabic language phishing samples would greatly enhance applicability in Bahrain and the wider GCC region where the number of phishing campaigns targeting Arabic speakers is on the rise.

Adaptive retraining pipeline: Concept drift over time reduces the accuracy of models when phishing methods are constantly changing (Gama et al., 2014). An automated pipeline that ingests new phishing samples on a periodic schedule could ensure that the classification would not need to be manual.

Email header analysis: Only body text of emails is analysed. Adding email header information such as SPF, DKIM and DMARC Header: Results, Reply-To, domain age, etc. mismatches, would greatly lessen false positives from vocabulary collisions between genuine and spam commercial emails and spam vocabulary.

Real-time analyst dashboard: The Streamlit-based dashboard could enable security analysts to submit queries, visually explore phishing probability distributions within email batches and review flagged emails via a structured workflow, moving beyond a command-line prototype to a deployable security operations tool.

Multi-agent architecture: A sub-agent design of these specialists would allow for the separation of classification, explanation, and guidance sub-agents, which may lead to better response quality and the ability to execute the tools parallelly, thus reducing the response time.

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AI Disclosure

This project has been done with the support from AI tools and following Bahrain Polytechnic Academic integrity guidelines. The following tool was used:

For code generation, debugging, system design guidance, referencing in APA style and summarizing research papers Claude (Anthropic) helped with assistance throughout the project. All code that was created for the student was reviewed, tested and validated against actual Azure deployments. Student configured, deployed and tested all Azure services. This report contains all the evaluation results as actual output from the deployed system on the student's Azure subscription (IT9204).

The system design, system architecture and architectural decisions, technologies used, evaluation methodology, and the analysis of results are the student's own work. The code provided, policy, and configuration documents were written and tested by the student with AI assistance.

Project Resources

Resource	Link
Dataset	https://www.kaggle.com/datasets/subhajournal/phishingemails
Demo Video	https://youtu.be/FgLfeEq4M-k
Source Code	https://github.com/Fatimaxx24/202508958-IT9204-Emerging-Technologies-in-AI

Table 13 - Project Resources